

half of the time for countries including Spain, Germany, France, Italy, and Austria. Italians attempting to use the 4 percent rule in their domestic financial markets would have actually faced failure in 76 percent of the historical periods (24 percent success). The exhibit also shows results for a 5 percent withdrawal rate, and failure rates are substantially higher than for 4 percent. Even if the 4 percent rule could somehow be deemed safe, there is clearly not much room for error when seeing how quickly failure rates rise for 5 percent withdrawals. It is somewhat of an understatement to suggest that the results in this exhibit do not portray the 4 percent rule in a positive light.

Keep in mind also that I am only looking at the twenty developed-market countries in the data set, with data going back to 1900. Travel back in time to 1900, though, and ask people to put together a list of twenty developed-market countries for the twentieth century, and you would probably find frequent mention of countries like Argentina, Russia, and China, among others. As those countries never made the data set, even the results I describe here include a degree of survivorship bias.

Thus far, the international data set has revealed sustainable spending rates that fall well below what has been observed with only US historical data. Exhibit 5.3 provides further insight into how the SAFEMAX is impacted by asset allocation. For each stock allocation, the perfect foresight assumption is allowed to guide an optimal asset allocation for bonds and bills, which allows for outcomes slightly different than what was seen in Exhibit 5.2 when only bills were used. While withdrawal rates were often less, this exhibit does, nonetheless, support Bengen's original argument in favor of aggressive asset allocations. For eighteen of the twenty countries, as well as the world and world ex-US, the highest SAFEMAX was obtained with at least 50 percent stocks. Only Switzerland (20 percent stocks) and Sweden (40 percent stocks) witnessed the highest SAFEMAX with less than 50 percent stocks. Especially for the countries with the lowest sustainable spending rates, aggressive stock allocations helped. The seven countries with the lowest SAFEMAX could achieve these only with stock allocations of 80 percent or higher. Because of inflation, nominal bonds were even riskier than stocks for retirement portfolios in those countries.

Exhibit 5.3

Maximum Sustainable Withdrawal Rate for Thirty Years

By Percentage Allocation to Stocks, Combined with the Optimal Bond/Bill Allocation for the Remainder